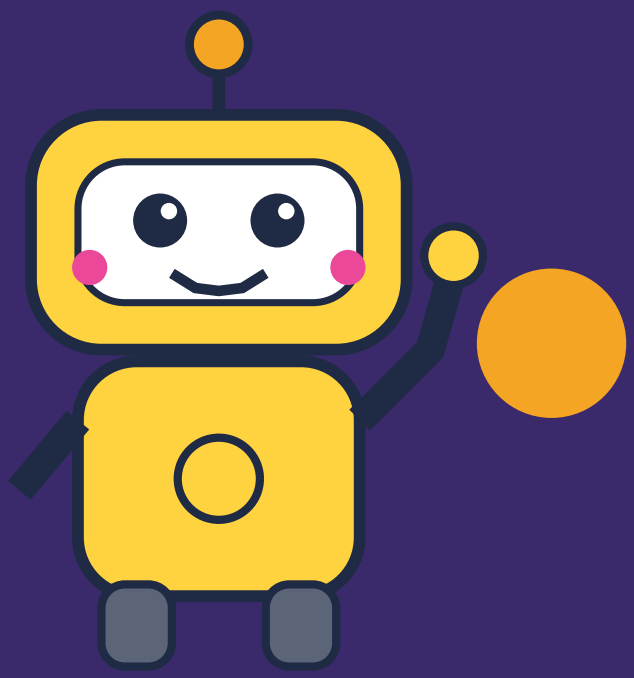
ASSESSMENT BOOK

Think Like a Coder

Chapter assessments for Class 6

14 one-page assessments, one for every chapter
10 marks each: choose, apply and think like a coder
Answers, marking guides and a class record sheet



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 6 · Photocopiable for classroom use

SAMPLE EDITION · DRAFT

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Using these assessments

This book has one assessment for every chapter of *Think Like a Coder*, Class 6. Each assessment fits on one page, takes about 30 minutes and is marked out of 10. Use it after the chapter's last period, or as a revision check later in the term. All questions are new: they test the same skills as the workbook, not the same questions.

Section	What it checks	Questions	Marks
A	Knowing and understanding the chapter's ideas	3 multiple choice	3 x 1
B	Applying the idea to a short problem, with working	2 short answers	2 x 2
C	Thinking like a coder: write, trace, fix or explain a program or plan	1 task	1 x 3

Turning a score into a journal level

Score	Journal level	What it usually means
9-10	Extending	Solves new problems and explains the reasoning
7-8	Secure	Applies the idea correctly on their own
4-6	Developing	Understands with prompts; makes some slips
0-3	Beginning	Needs more practice with the chapter's idea

✓ Good practice

- ✓ Read the questions aloud for children who find reading hard: we are testing thinking, not reading.
- ✓ Give full marks for any correct method, even if it differs from the answer key; the marking guides list what to look for.
- ✓ Give part marks in Sections B and C for correct working, even when the final answer has a slip.
- ✓ Record the score on the class record sheet, and use it with your observations in the journal, not instead of them.
- ✓ Go over the test with the class: the most common mistakes are a lesson plan for the next period.

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Assessment 1 Patterns in Mathematics

Coder skill: Rules and functions: name a rule and reuse it · Level C1 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 What is the next term? 4, 7, 10, 13, ... [1 mark]

a) 15 **b)** 16 **c)** 17

2 square(5) gives back ... [1 mark]

a) 10 **b)** 25 **c)** 55

3 triangle(5) = 1 + 2 + 3 + 4 + 5 = ... [1 mark]

a) 10 **b)** 15 **c)** 25

Section B Answer in a few words or steps — 4 marks

4 The rule is term = 3 x position + 2. Find the 10th term and the 100th term. [2 marks]

5 What do the first 6 odd numbers add up to? How can you tell without adding? [2 marks]

Section C Think like a coder — 3 marks

6 The sequence is 5, 9, 13, 17, ... Write a 'next term' rule and a 'position' rule. Use the position rule to find the 50th term. [3 marks]

How did I do? Colour a face.

I can finish this test.	I can show my thinking.	I can check my answers.
		

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Assessment 2 Lines and Angles

Coder skill: Drawing programs: angles, turns and a square(size) function · Level C1 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 An angle of 120° is ... [1 mark]

a) acute **b)** obtuse **c)** reflex

2 On an 8-point compass, neighbouring directions are ... apart. [1 mark]

a) 30° **b)** 45° **c)** 90°

3 A robot draws a regular pentagon. Each turn is ... [1 mark]

a) 60° **b)** 72° **c)** 108°

Section B Answer in a few words or steps — 4 marks

4 (a) Face NE and turn 90° clockwise. (b) Face S and turn 135° anticlockwise. Where do you face each time? [2 marks]

5 Write a function square(size) that draws a square of any size. [2 marks]

Section C Think like a coder — 3 marks

6 Write a function polygon(sides, size) that draws any regular polygon. Show the call that draws a hexagon with sides of 40, and the turn it uses. [3 marks]

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Assessment 3 Number Play

Coder skill: Algorithms that repeat until something happens · Level C1 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Which number is a palindrome? [1 mark]

a) 345 **b)** 373 **c)** 337

2 With a 'repeat until' loop ... [1 mark]

a) you may not know the number of steps in advance

b) it always runs 10 times

c) it never stops

3 Reverse and add once, starting with 23. You get ... [1 mark]

a) 55 **b)** 46 **c)** 32

Section B Answer in a few words or steps — 4 marks

4 Reverse and add until you get a palindrome, starting with 68. Write each step. [2 marks]

5 Rule: if the number is even, halve it, else multiply by 3 and add 1. Repeat until you reach 1. Start at 6 and write every number. [2 marks]

Section C Think like a coder — 3 marks

6 Write an algorithm with 'repeat until' that adds the digits of a number again and again until one digit is left. Trace it for 987. [3 marks]

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Assessment 4 Data Handling and Presentation

Coder skill: Lists: store, sort and find the biggest · Level C1 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 marks = [12, 8, 15, 9]. Which item is at position 3? [1 mark]

a) 8 **b)** 15 **c)** 9

2 Scale: 1 square = 5 students. A bar is 7 squares tall. It shows ... [1 mark]

a) 7 students **b)** 12 students **c)** 35 students

3 Every graph needs a title, labels and a ... [1 mark]

a) scale **b)** colour **c)** password

Section B Answer in a few words or steps — 4 marks

4 Find the biggest: set biggest to the first item; for each item, if item > biggest then biggest = item. The list is 14, 9, 21, 17, 25, 6. Write biggest after each item. [2 marks]

5 Votes: tea, milk, juice, milk, tea, milk, juice, milk. Make a frequency table. [2 marks]

Section C Think like a coder — 3 marks

6 Books read: 6A 45, 6B 30, 6C 55, 6D 20. Choose a scale for a pictograph and say how many symbols each class needs. How many books in all? [3 marks]

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Assessment 5 Prime Time

Coder skill: The sieve and a factor-finding algorithm · Level C2 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Which number is prime? [1 mark]

a) 21 **b)** 27 **c)** 29 **d)** 33

2 Which is a factor of 36? [1 mark]

a) 7 **b)** 8 **c)** 9 **d)** 10

3 How many factors does a prime number have? [1 mark]

a) 1 **b)** 2 **c)** 3

Section B Answer in a few words or steps — 4 marks

4 List all the factors of 30. [2 marks]

5 In the sieve for 1 to 50, after crossing out multiples of 2, 3, 5 and 7 you can stop. Why? [2 marks]

Section C Think like a coder — 3 marks

6 Algorithm: for d = 1, 2, 3, ... while d x d is not more than n: if d divides n, write d and n ÷ d. Trace it for n = 48. How many factors does 48 have? [3 marks]

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Assessment 6 Perimeter and Area

Coder skill: Functions with two inputs; breaking shapes apart · Level C2 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 The area of a rectangle 7 cm by 4 cm is ... [1 mark]

a) 11 cm² **b)** 22 cm² **c)** 28 cm²

2 The perimeter of the same rectangle is ... [1 mark]

a) 11 cm **b)** 22 cm **c)** 28 cm

3 The function area(l, w) has how many inputs? [1 mark]

a) 1 **b)** 2 **c)** 3

Section B Answer in a few words or steps — 4 marks

4 A square has an area of 49 cm². What is its side, and its perimeter? [2 marks]

5 perimeter(l, w) = 2 x (l + w). Find perimeter(9, 6). Give another rectangle with the same perimeter. [2 marks]

Section C Think like a coder — 3 marks

6 An L-shaped room is made of rectangle A, 10 m by 4 m, and rectangle B, 4 m by 3 m, joined at one end. Use area(l, w) to find the area of the room. [3 marks]

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Assessment 7 Fractions

Coder skill: Compare, simplify and sort fractions · Level C2 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Which fraction is equal to 2/3? [1 mark]

a) 4/6 **b)** 3/4 **c)** 2/6

2 12/18 in simplest form is ... [1 mark]

a) 6/9 **b)** 2/3 **c)** 4/6

3 Bubble sort compares ... [1 mark]

a) neighbours **b)** the first and last items

c) random items

Section B Answer in a few words or steps — 4 marks

4 Order from smallest to biggest: 3/4, 2/3, 5/6. [2 marks]

5 Hindi: 36/40. English: 45/50. In which subject did Sam do better? [2 marks]

Section C Think like a coder — 3 marks

6 Do one bubble-sort pass on 5, 2, 4, 1 (smallest first). Write the list after each comparison. [3 marks]

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Assessment 8 Playing with Constructions

Coder skill: Construction scripts: exact steps and debugging · Level C2 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 A construction uses only ... [1 mark]

a) a ruler and a compass **b)** a protractor and a ruler

c) a calculator

2 All the points the same distance from a centre make a ... [1 mark]

a) circle **b)** square **c)** line

3 The perpendicular bisector of AB cuts AB at its ... [1 mark]

a) midpoint **b)** end **c)** point A

Section B Answer in a few words or steps — 4 marks

4 AB = 6 cm. A script draws arcs of radius 2 cm from A and from B, but they do not meet. Find the bug and fix it. [2 marks]

5 AB = 5 cm. Point P is 4 cm from A and 4 cm from B. How can you find P with a compass? [2 marks]

Section C Think like a coder — 3 marks

6 Write a script to construct a triangle with sides 5 cm, 4 cm and 3 cm. [3 marks]

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Assessment 9 Symmetry

Coder skill: Reflect and rotate: functions that change shapes · Level C2 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 How many lines of symmetry does an equilateral triangle have? [1 mark]

a) 1 **b)** 3 **c)** 6

2 What is the order of rotational symmetry of a rectangle that is not a square? [1 mark]

a) 1 **b)** 2 **c)** 4

3 Which capital letter has rotational symmetry of order 2? [1 mark]

a) S **b)** A **c)** E

Section B Answer in a few words or steps — 4 marks

4 Columns A to H; the mirror line is between D and E. Where do the squares B3 and D5 go when reflected? [2 marks]

5 How many lines of symmetry does a regular hexagon have? What is its order of rotational symmetry? [2 marks]

Section C Think like a coder — 3 marks

6 Columns are numbered 1 to 8, with the mirror between 4 and 5. Write a rule for reflect: new column = ? Use it for columns 1, 3 and 4. [3 marks]

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Assessment 10 The Other Side of Zero

Coder skill: Integers in data: lowest, highest and range · Level C2 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Which is the smallest? [1 mark]

a) -3 **b)** -7 **c)** 0 **d)** 2

2 -4 + 6 = ... [1 mark]

a) 2 **b)** -10 **c)** 10

3 3 - (-5) = ... [1 mark]

a) -2 **b)** 8 **c)** -8

Section B Answer in a few words or steps — 4 marks

4 A lift is at floor 2. It goes down 5 floors, then up 1. Trace the floor after each move. [2 marks]

5 Temperatures in Leh: -12°C, -5°C, 3°C, -8°C. Find the highest, the lowest and the range. [2 marks]

Section C Think like a coder — 3 marks

6 A bank balance starts at ₹500. Trace it: spend ₹700, earn ₹300, spend ₹250. Is it negative at the end? [3 marks]

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Assessment AI1 Introduction to AI

Coder skill: Tell AI from automation; how machines learn · Level C3 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Which one uses AI? [1 mark]

a) a washing-machine timer
b) face unlock that learnt your face
c) a fan regulator

2 Learning from labelled examples is called ... [1 mark]

a) supervised learning
b) unsupervised learning
c) reinforcement learning

3 Learning by trial, reward and penalty is called ... [1 mark]

a) supervised learning
b) unsupervised learning
c) reinforcement learning

Section B Answer in a few words or steps — 4 marks

4 AI or automation? (a) A street light that switches on at 6 pm every day. (b) A music app that suggests songs you might like. [2 marks]

5 A shop's computer groups customers by what they buy. Nobody tells it the groups. Which kind of learning is this? Why? [2 marks]

Section C Think like a coder — 3 marks

6 Explain how you would teach a computer to tell apples from oranges using supervised learning. [3 marks]

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Assessment AI2 Basic Data Concepts

Coder skill: Types of data; collect, organise and show it · Level C3 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Raw facts such as numbers, words and pictures are called ... [1 mark]

a) data **b)** information **c)** a chart

2 A survey is a way to ... [1 mark]

a) collect data **b)** store data **c)** delete data

3 A photo of a leaf is ... data. [1 mark]

a) number **b)** picture **c)** sound

Section B Answer in a few words or steps — 4 marks

4 Key: one tree symbol = 10 trees. How many symbols show 45 trees, and how many show 25 trees? [2 marks]

5 6A read 12 books and 6B read 18 books. Write one piece of data and one piece of information from this. [2 marks]

Section C Think like a coder — 3 marks

6 Plan how to answer: 'How do students in our school come to school?' Say how you will collect, organise and show the data. [3 marks]

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Assessment AI3 Patterns and Decisions

Coder skill: Learn a rule from labelled examples, then decide · Level C3 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 'It rained on 4 of the last 5 Mondays' is ... [1 mark]

a) an observation **b)** a conclusion **c)** a rule

2 A spam filter learns from ... [1 mark]

a) labelled examples **b)** random guesses

c) coloured emails

3 A conclusion from very little data ... [1 mark]

a) may be wrong **b)** is always right

c) cannot be tested

Section B Answer in a few words or steps — 4 marks

4 Labelled examples: barks, 4 legs: dog. Meows, 4 legs: cat. Barks, 3 legs: dog. Meows, 4 legs: cat. Which feature decides the label, and which does not? [2 marks]

5 Use the rule you found. Predict the label for (a) barks, 4 legs and (b) meows, 3 legs. [2 marks]

Section C Think like a coder — 3 marks

6 A rule was learnt from 5 examples: 'All birds can fly.' Give an example that breaks it. Explain how more and fairer data would help. [3 marks]

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Assessment AI4 Ethics and Digital Responsibility

Coder skill: Strong passwords, privacy and digital footprints · Level C3 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Which password is the strongest? [1 mark]

a) password123 **b)** Riya2014

c) Kite!Mango7River

2 Tricking people into giving away their passwords is called ... [1 mark]

a) phishing **b)** piracy **c)** plagiarism

3 Location data collected without you noticing is a ... footprint. [1 mark]

a) active **b)** passive

Section B Answer in a few words or steps — 4 marks

4 An email says: 'Your account will be closed today! Click here and type your password.' Give two warning signs and say what you should do. [2 marks]

5 Active or passive footprint? (a) Posting a photo. (b) A website saving cookies about your visits. [2 marks]

Section C Think like a coder — 3 marks

6 Make up a strong password with the three-word method (not one you really use). Then write two rules for keeping passwords safe. [3 marks]

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Answers and marking guide

Accept any correct method. Part marks are shown in brackets for Sections B and C.

1. Patterns in Mathematics

1. b) 16
2. b) 25
3. b) 15
4. 32 and 302 (1 mark for each)
5. 36, because the first n odd numbers add up to n x n (6 x 6) (1 mark for 36, 1 mark for the reason)
6. Next: add 4. Position: term = 4 x position + 1. 50th term = 201. (1 mark for each)

2. Lines and Angles

1. b) obtuse
2. b) 45°
3. b) 72°
4. (a) SE (b) NE (1 mark for each)
5. define square(size): repeat 4 times: forward size, turn right 90° (1 mark for using the input size, 1 mark for the loop with 90° turns)
6. define polygon(sides, size): repeat sides times: forward size, turn right 360 ÷ sides. Call: polygon(6, 40). Turn: 60°. (1 mark for the function, 1 mark for the call, 1 mark for 60°)

3. Number Play

1. b) 373
2. a) you may not know the number of steps in advance
3. a) 55
4. 68 + 86 = 154; 154 + 451 = 605; 605 + 506 = 1111 (2 marks; 1 mark if one step has a slip)
5. 6, 3, 10, 5, 16, 8, 4, 2, 1 (2 marks; 1 mark with one slip)
6. Repeat until the number has one digit: number = sum of its digits. 987: 9 + 8 + 7 = 24; 2 + 4 = 6. (1 mark for the loop, 1 mark for the stop condition, 1 mark for the trace)

4. Data Handling and Presentation

1. b) 15
2. c) 35 students
3. a) scale
4. 14, 14, 21, 21, 25, 25 (2 marks; 1 mark with one slip)
5. tea 2, milk 4, juice 2 (2 marks; 1 mark with one slip)
6. For example, 1 symbol = 10 books: 6A 4½, 6B 3, 6C 5½, 6D 2. Total 150 books. (1 mark for a sensible scale, 1 mark for the symbols, 1 mark for 150)

5. Prime Time

1. c) 29
2. c) 9
3. b) 2
4. 1, 2, 3, 5, 6, 10, 15, 30 (2 marks; 1 mark if one is missing)
5. The next prime is 11, and 11 x 11 = 121 is past 50 (2 marks; 1 mark for a partial reason)
6. 1 & 48, 2 & 24, 3 & 16, 4 & 12, 6 & 8; stop at d = 7 (49 > 48). 48 has 10 factors. (1 mark for the pairs, 1 mark for the stop, 1 mark for 10)

6. Perimeter and Area

1. c) 28 cm^2
2. b) 22 cm
3. b) 2
4. 7 cm; 28 cm (1 mark for each)
5. 30; for example 10 by 5 (1 mark for each)
6. area(10, 4) = 40; area(4, 3) = 12; total 52 m^2 (1 mark for each area, 1 mark for the total)

7. Fractions

1. a) 4/6
2. b) 2/3
3. a) neighbours
4. 2/3, 3/4, 5/6 (2 marks; 1 mark for writing them out of 12)
5. Neither: both are 90 out of 100 (1 mark for changing to 100, 1 mark for 'equal')
6. Swap 5, 2: 2, 5, 4, 1. Swap 5, 4: 2, 4, 5, 1. Swap 5, 1: 2, 4, 1, 5. (1 mark for each comparison)

8. Playing with Constructions

1. a) a ruler and a compass
2. a) circle
3. a) midpoint
4. The radius must be more than half of AB (3 cm). Use, for example, 4 cm. (1 mark for the bug, 1 mark for the fix)
5. Draw arcs of radius 4 cm from A and from B; P is where they cross (1 mark for each step)
6. 1. Draw AB = 5 cm. 2. With the compass at A, draw an arc of radius 4 cm. 3. With the compass at B, draw an arc of radius 3 cm to cross it at C. Join AC and BC. (1 mark for each clear step)

9. Symmetry

1. b) 3
2. b) 2
3. a) S
4. G3 and E5 (1 mark for each)
5. 6 and 6 (1 mark for each)
6. new column = 9 - column; 1 goes to 8, 3 to 6, 4 to 5 (1 mark for the rule, 2 marks for all three (1 mark for two))

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10. The Other Side of Zero

1. b) -7
2. a) 2
3. b) 8
4. -3, then -2 (1 mark for each)
5. highest 3°C, lowest -12°C, range 15°C (2 marks; 1 mark for highest and lowest only)
6. -200, then 100, then -150. Yes, it ends at -₹150. (1 mark for each step)

AI 1. Introduction to AI

1. b) face unlock that learnt your face
2. a) supervised learning
3. c) reinforcement learning
4. (a) automation (b) AI (1 mark for each)
5. Unsupervised learning, because there are no labels and it finds the groups on its own (1 mark for the type, 1 mark for the reason)
6. Collect many photos of apples and oranges; label each one; let the computer learn from them; test it on new photos it has not seen. (1 mark for data, 1 mark for labels, 1 mark for testing)

AI 2. Basic Data Concepts

1. a) data
2. a) collect data
3. b) picture
4. 4½ and 2½ (1 mark for each)
5. Data: 6B read 18 books. Information: 6B read 6 more books than 6A. (1 mark for each)
6. Collect: a survey of each class. Organise: a table with walk, cycle, bus, car and a count for each. Show: a bar chart. (1 mark for each stage)

AI 3. Patterns and Decisions

1. a) an observation
2. a) labelled examples
3. a) may be wrong
4. The sound (barks or meows) decides; the number of legs does not (1 mark for each)
5. (a) dog (b) cat (1 mark for each)
6. A penguin or an ostrich cannot fly. More and varied examples, including birds that cannot fly, give a better rule; then test it on new birds. (1 mark for the example, 1 mark for more and varied data, 1 mark for testing)

AI 4. Ethics and Digital Responsibility

1. c) Kite!Mango7River
2. a) phishing
3. b) passive
4. It rushes you and asks for your password through a link. Do not click; tell a trusted adult. (1 mark for two signs, 1 mark for the action)
5. (a) active (b) passive (1 mark for each)
6. For example: Lamp#Tiger4Cloud. Rules: never share it; use a different password for each account; do not use your name or birthday. (1 mark for a strong password, 1 mark for each rule)

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Class record sheet

Class 6 ____ Section ____ Term ____. Write each score out of 10. You may photocopy this page.

Student	1	2	3	4	5	6	7	8	9	10	AI1	AI2	AI3	AI4	Avg

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